Data Structures and Algorithms (DSA)

Xonix Enhanced: Powered by Data Structures

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Contents

Introduction:

Overview:

We improved the XONIX game by providing a real game like experience. By having just a single multiplayer code we added our creativity and implemented key data structures showcasing our strong concepts.

Features:

We implemented the following features in our version of XONIX game:

1. Main Menu
2. Login and Registration
3. Authentication
4. Friend System
5. Matchmaking Queue
6. Leaderboard
7. Inventory System
8. Multiplayer Mode
9. Player Profile
10. Save and Load game

Tools & Programming Languages:

The project was done on Visual Studio 2022. The game is a pure version of C++ (for coding and logic) and SFML (for aesthetics and visuals). CSV files were used to store player data, themes and game state for saving and loading.

Work Distribution:

The following is the work distribution of both members. The work includes implementations of both code and visuals.

Talha Qamar	Hassan Abid
Login and Sign Up Pages with authentication.	Sprites and Aesthetics.
Single player and Multiplayer Pages.	Main Menu.
Point system and powerups.	Multiplayer.
Leaderboard and Inventory page.	Friend System.
Save Game.	Load Game.

Data Structures Used:

AVL Tree:

The AVL tree was used for storing a structure of themes. Traversing the tree provided themes from which user can choose based on their liking. AVL was used for its log(n) traversal and storage based on sorting.

Min-Heap:

The Min-Heap was used to display leaderboard of players based on high score. The minheap was traversed in reverse order to display highest to lowest. Min-Heap was used as it stores in an ascending order.

Linked List:

Linked List was used for friends-based system. It includes current friends and friend requests stored in a LinkedList. LinkedList provides easy addition and removal without resizing.

Hash Table:

The hash table was used for constant time search to find and create friends. Constant time search was the key factor of using Hash Table.

Queue:

A priority-based queue was used for efficient and fair matchmaking. This allowed players of same caliber to compete in a multiplayer game. A priority queue stores sorted data based on specific parameters. In this case it allowed better, quick and fair matchmaking.

Workflow and Implementation Timeline:

We started with a planned workflow to ensure a smooth project experience.

Planning and Development:

We started of with selecting a theme to go with. Decided sprites and themes to go with to maintain consistency. After that we worked on authentication which consists of login and sign up pages. Moving forward we made all the necessary pages and buttons that would connect features. After the front end we started the backend which included the implementation of data structures and game implementations. Powerups and points system were added. Then leaderboard and inventory pages were implemented. This was followed by friend’s system and multiplayer. In the end, save game and load game was implemented.

Challenges and Solution:

Challenges:

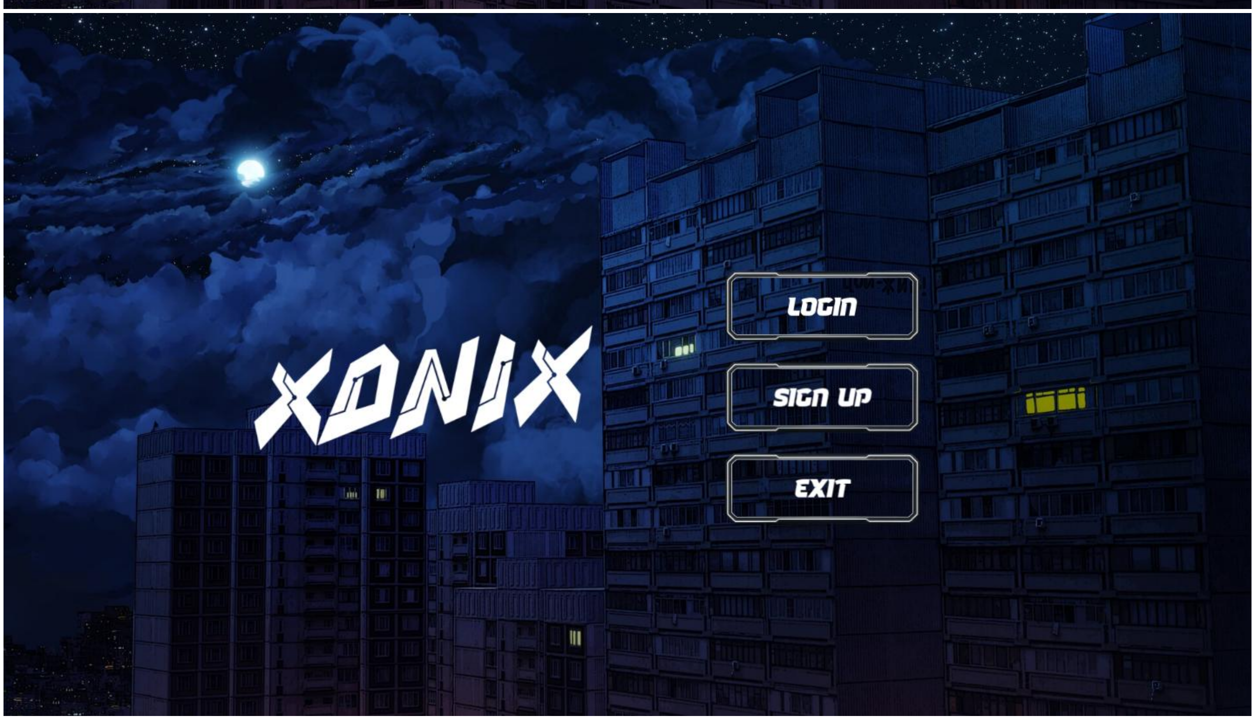
1. The friend system had a small but potent issue. A shallow copy was being created which costed us 1-2 days of our project.
2. Many header files and different modules made merging a challenge.
3. The war situation affected our workflow and coordination.

Solutions:

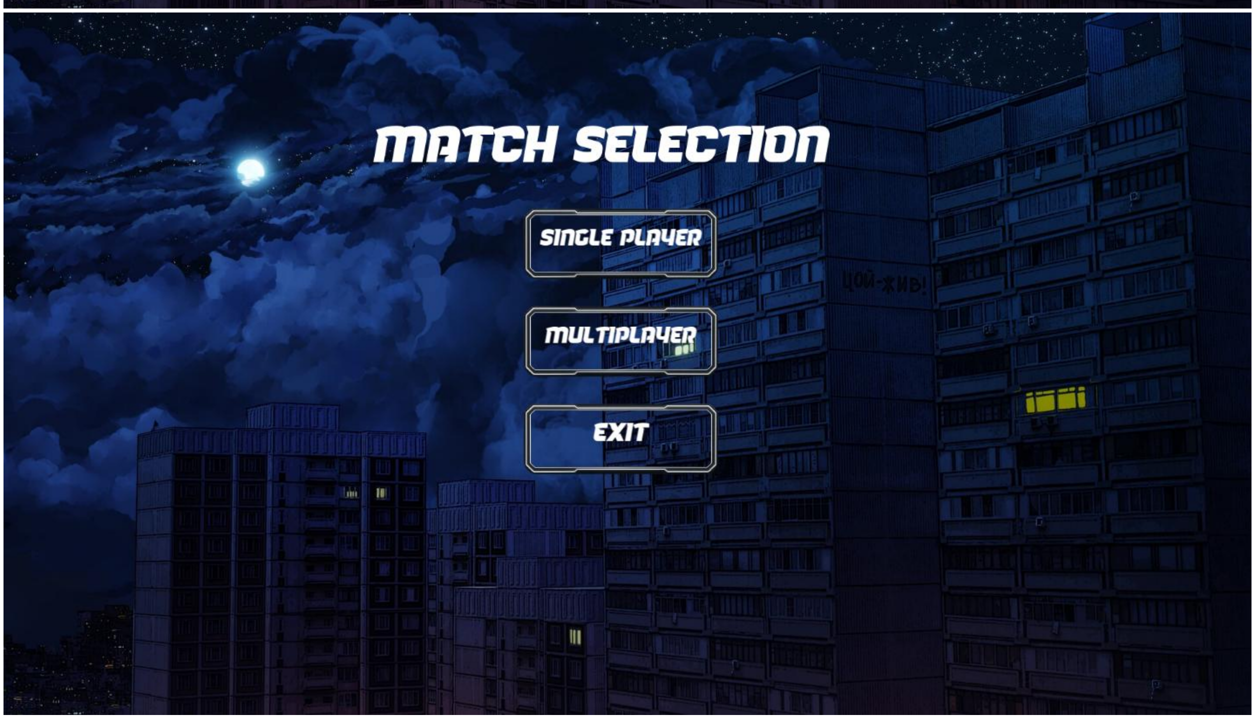
1. A copy constructor solved this problem.
2. We merged feature by feature to avoid errors and uncertainties.
3. An early ceasefire saved us from wasting more time and focusing on project solely.

Screenshots and Sample Outputs:

The following are images of our game in order showcasing our creativity and concepts visually:



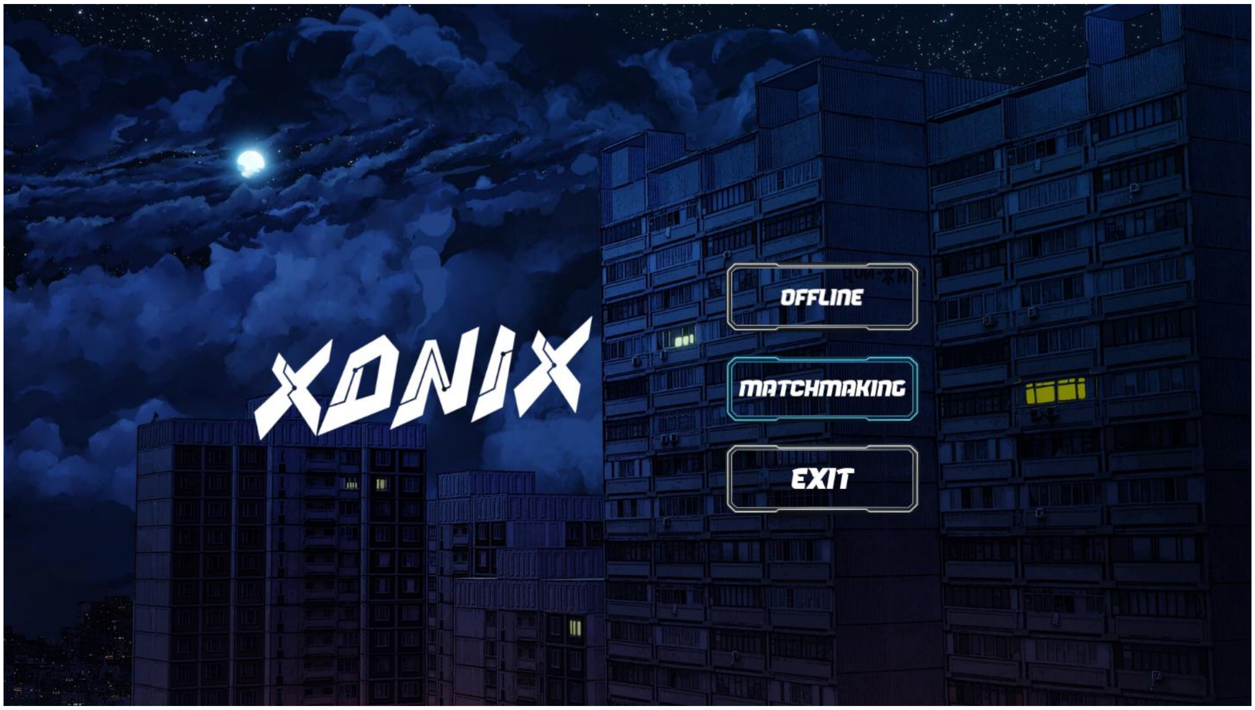

XONIX
USERNAME:
hello
PASSWORD:
***
NICKNAME:
Please fill in all fields.
SIGN UP

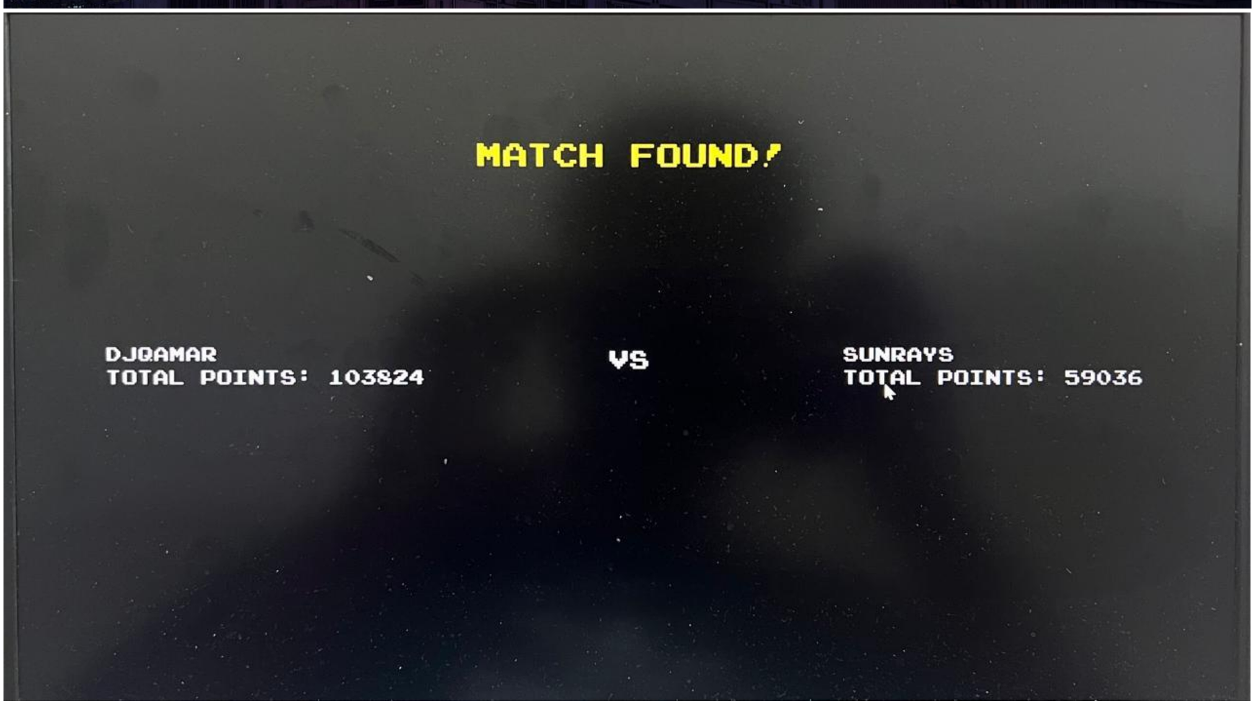

MATCH SELECTION
SINGLE PLAYER
MULTIPLAYER
EXIT

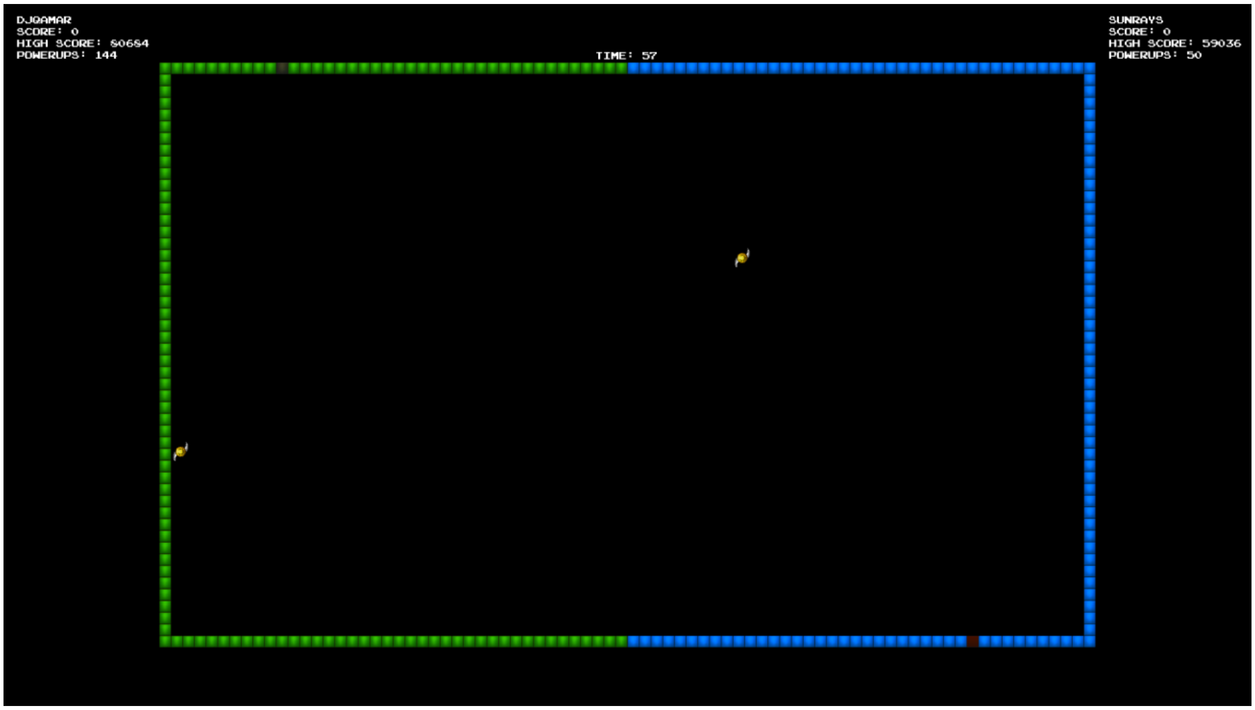

DJQAMAR
RANK :
FRIENDS LOG OUT
START
LEADERBOARD
INVENTORY
EXIT
XONIX


LEADERBOARD
RANK NICKNAME HIGH SCORE
1 DJQAMAR 80684
2 SUNRAYS 59036
3 G4ATT 14976
4 BOHOT 2592
5 JUTT 80
XONIX

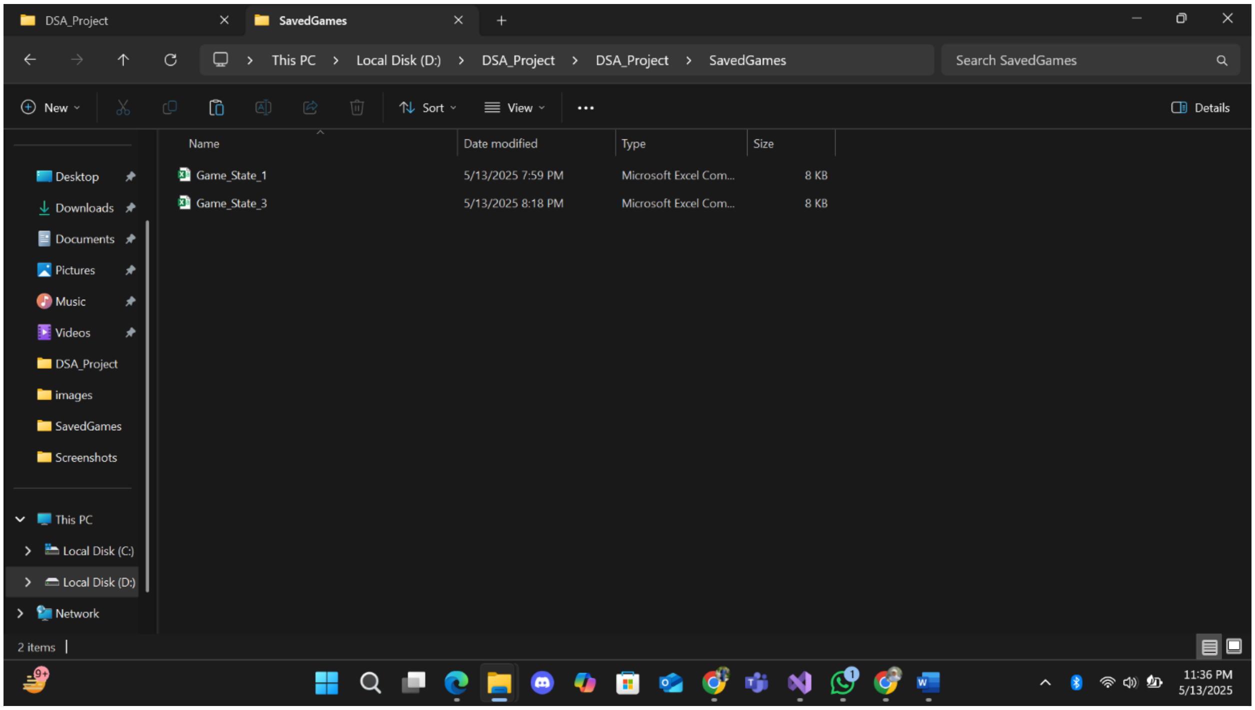
Multiplayer:

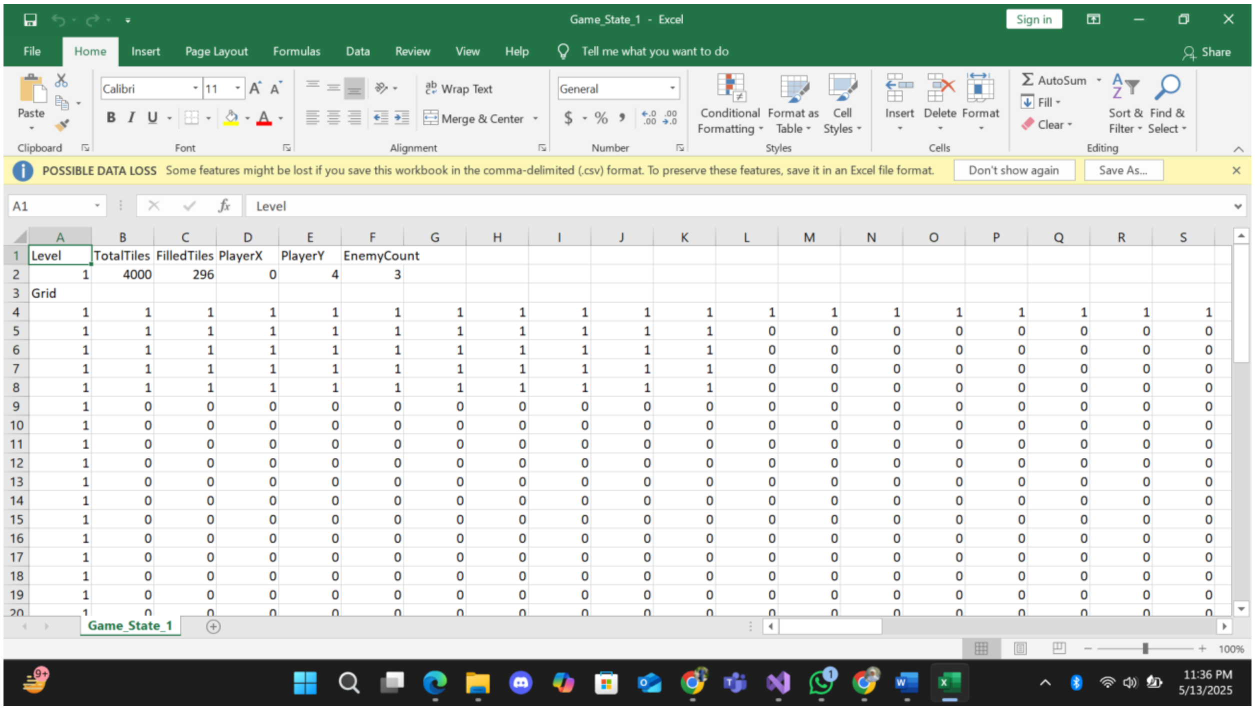

XONIX
OFFLINE
MATCHMAKING
EXIT


MATCH FOUND!
DJQAMAR
TOTAL POINTS: 103824
VS
SUNRAYS
TOTAL POINTS: 59036



File Handling:

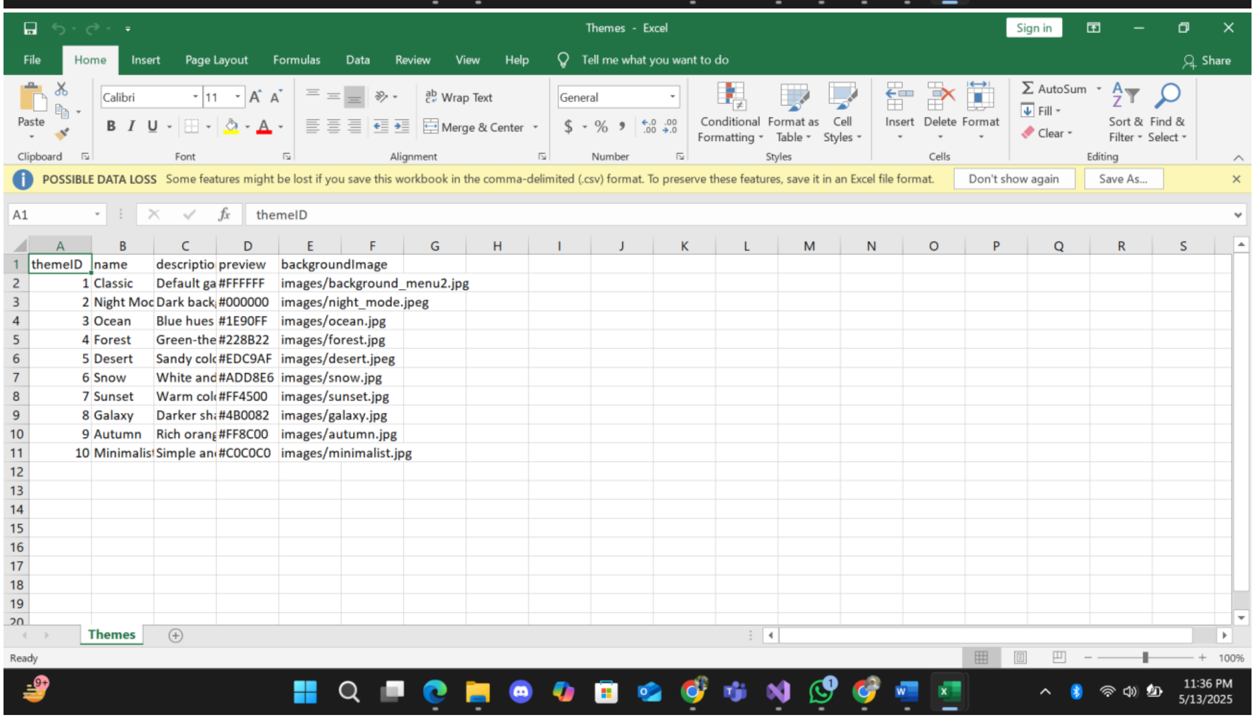




Game_State_1 - Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Level	TotalTiles	FilledTiles	PlayerX	PlayerY	EnemyCount													
2	1	4000	296	0	4	3													
3	Grid																		
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
6	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
7	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
8	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
9	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
17	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

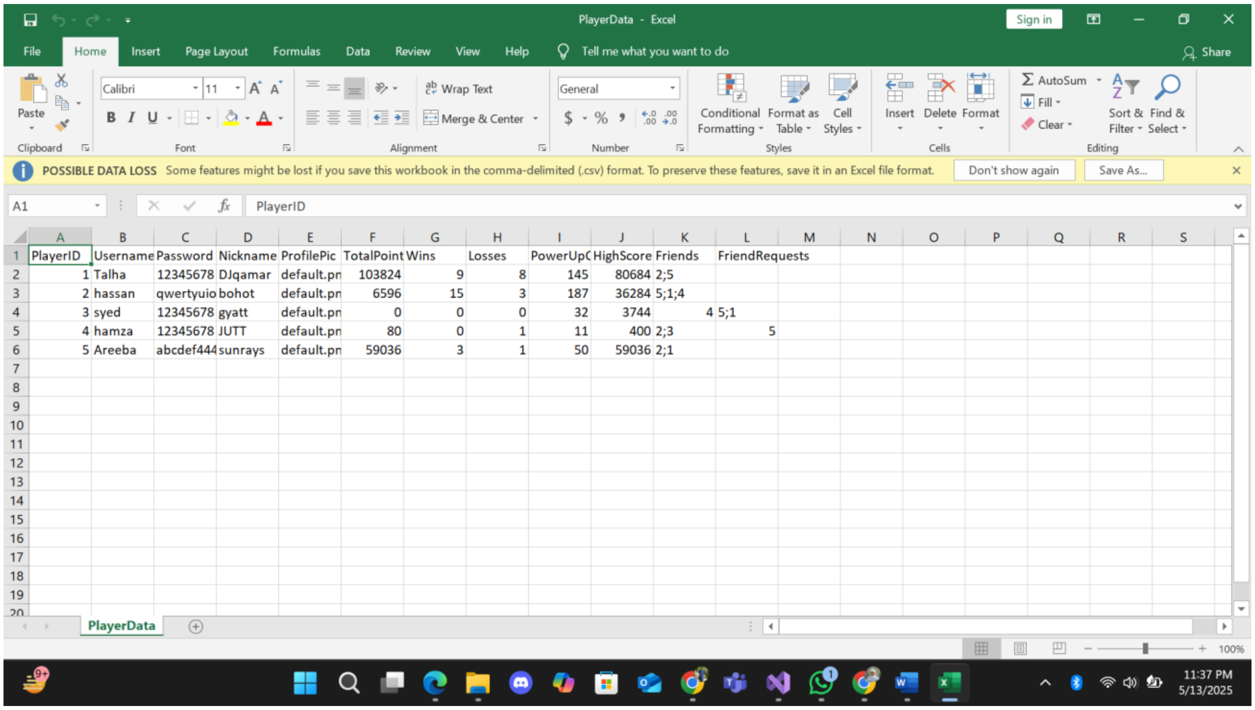
Game_State_1



Themes - Excel

	A	B	C	D	E
1	themeID	name	descriptio	preview	backgroundImage
2	1	Classic	Default ga	#FFFFFF	images/background_menu2.jpg
3	2	Night Moc	Dark back	#000000	images/night_mode.jpeg
4	3	Ocean	Blue hues	#1E90FF	images/ocean.jpg
5	4	Forest	Green-the	#228B22	images/forest.jpg
6	5	Desert	Sandy colc	#EDC9AF	images/desert.jpeg
7	6	Snow	White and	#ADD8E6	images/snow.jpg
8	7	Sunset	Warm colc	#FF4500	images/sunset.jpg
9	8	Galaxy	Darker sh	#4B0082	images/galaxy.jpg
10	9	Autumn	Rich orang	#FF8C00	images/autumn.jpg
11	10	Minimalist	Simple an	#C0C0C0	images/minimalist.jpg

Themes



PlayerID	Username	Password	Nickname	ProfilePic	TotalPoint	Wins	Losses	PowerUpC	HighScore	Friends	FriendRequests
1	Talha	12345678	DJqamar	default.pn	103824	9	8	145	80684	2;5	
2	hassan	qwertyuio	bohot	default.pn	6596	15	3	187	36284	5;1;4	
3	syed	12345678	gyatt	default.pn	0	0	0	32	3744	4	5;1
4	hamza	12345678	JUTT	default.pn	80	0	1	11	400	2;3	5
5	Areeba	abcdef444	sunrays	default.pn	59036	3	1	50	59036	2;1	

<u>Conclusion:</u>

We learned a lot in terms of coding, game development and most importantly data handling. The concepts were cleared much more which allowed a lot of practical insights.

What we would work on is to manage time better and break into parts. This would allow better management. Another important thing is to be focused at all times and try to set earlier deadlines to cover for unforeseen circumstances.